

Decimal Operations

Every graded set, every week — 5 weeks, 25 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Adding Decimals

Line up the points, fill with zeros, then add as normal.

Warm-Up 12 ITEMS • RECALL

1 $0.3 + 0.4$

2 $1.2 + 2.5$

3 $0.25 + 0.25$

4 $3.5 + 1.5$

5 $0.6 + 0.9$

6 $2.75 + 1.25$

7 $0.5 + 0.2$

8 $1.4 + 2.3$

9 $0.35 + 0.15$

10 $3.5 + 2.5$

11 $0.8 + 0.7$

12 $4.2 + 3.1$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $3.5 + 0.47$

14. $12.6 + 4.85$

15. $0.125 + 0.875$

16. $9.4 + 0.68$

17. $\$14.75 + \8.60 — type the number

18. $4.6 + 0.38$

19. $15.4 + 3.75$

20. $8.3 + 0.79$

21. $0.625 + 0.375$

22. $6.04 + 3.8$



Working space



Error journal

Subtracting Decimals

Same alignment. Fill the gaps with zeros before you borrow.

Warm-Up 12 ITEMS • RECALL

1 $0.9 - 0.4$

2 $3.7 - 1.2$

3 $1 - 0.5$

4 $2.5 - 0.5$

5 $0.75 - 0.25$

6 $5.5 - 2.5$

7 $0.8 - 0.5$

8 $4.7 - 2.2$

9 $1 - 0.4$

10 $3.5 - 1.5$

11 $0.85 - 0.35$

12 $6.9 - 3.4$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $4 - 1.35$

14. $10 - 0.07$

15. $8.2 - 3.45$

16. $\$20 - \13.68 — type the number

17. $6.05 - 2.5$

18. $5 - 2.45$

19. $10 - 0.09$

20. $9.2 - 4.55$

21. $7.05 - 3.5$

22. $13.4 - 5.85$

Multiplying Decimals

Multiply the digits, then count the decimal places.

Warm-Up 12 ITEMS • RECALL

1 0.5×4

2 0.2×6

3 1.5×2

4 0.25×4

5 3×0.1

6 0.5×10

7 0.5×6

8 0.2×8

9 1.5×4

10 0.25×8

11 2.5×2

12 0.1×9

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 0.4×0.2 (Two decimal digits in, two out)

14. 1.2×0.5

15. 2.5×1.4

16. 0.06×40

17. 3.2×2.5

18. 0.3×0.4

19. 1.5×0.6

20. 3.5×1.2

21. 0.08×50

22. 2.4×3.5

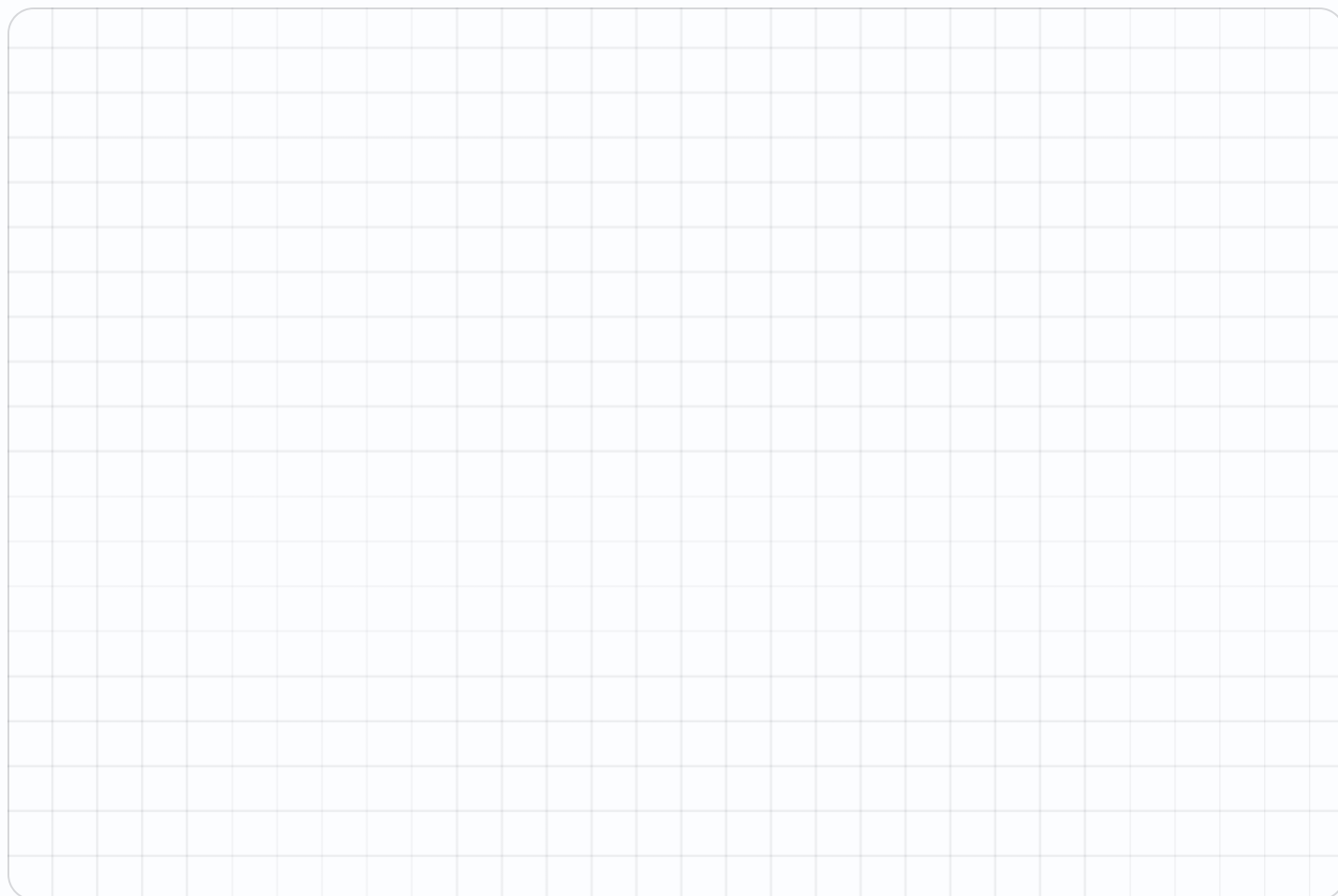
★ Challenge NOT GRADED • REASON IT OUT

23. 0.125×8
.....

24. 1.25×0.8
.....

25. 0.375×8
.....

26. 1.25×2.4
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Dividing Decimals

Shift both numbers the same number of places, then divide as usual.

Warm-Up 12 ITEMS • RECALL

1 $4.8 \div 2$

2 $0.9 \div 3$

3 $6.4 \div 4$

4 $2.5 \div 5$

5 $0.36 \div 6$

6 $7.2 \div 8$

7 $6.4 \div 2$

8 $0.8 \div 4$

9 $4.5 \div 5$

10 $8 \div 0.5$

11 $1.2 \div 3$

12 $6 \div 0.25$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $6 \div 0.5$ (How many halves in six)

14. $4 \div 0.25$

15. $1.5 \div 0.5$

16. $9.6 \div 1.2$

17. \$7.50 shared by 3 — type the number

18. $9.6 \div 8$

19. $1.28 \div 4$

20. $12 \div 0.4$

21. $7.2 \div 0.9$

22. $0.96 \div 1.2$



Unit Price Investigation

Enrichment. Price per unit is the only fair comparison.

✦ Warm-Up 4 ITEMS • RECALL

1 \$6 for 2 kg — price per kg

2 \$10 for 5 L — price per litre

3 \$9 for 3 — each

4 \$14 for 7 — each

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. \$4.50 for 3 kg — price per kg

.....

6. \$7.20 for 0.8 kg — price per kg

.....

7. \$5.40 for 4 — each

.....

8. \$8.70 for 6 — each

.....

Challenge

NOT GRADED • REASON IT OUT

9. Box A: \$5 for 400 g. Box B: \$9 for 800 g. Price per 100 g of B

10. Same boxes — price per 100 g of A

11. So which box is better value — type A or B

12. 600 g at \$7.20 — per 100 g

13. 1 kg at \$11 — per 100 g

14. Better value — type 600g or 1kg



Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE



Error journal

ONE SENTENCE PER MISTAKE

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 • Adding Decimals

OUT OF 22

Warm-Up • 1-12 • 0.7 | 3.7 | 0.5 | 5 | 1.5 | 4 | 0.7 | 3.7 | 0.5 | 6 | 1.5 | 7.3

Core • 13-22 • 3.97 | 17.45 | 1 | 10.08 | 23.35 | 4.98 | 19.15 | 9.09 | 1 | 9.84

Challenge • 23-26 • 6.375 | 5.4 | 7.875 | 6.3

1.2 • Subtracting Decimals

OUT OF 22

Warm-Up • 1-12 • 0.5 | 2.5 | 0.5 | 2 | 0.5 | 3 | 0.3 | 2.5 | 0.6 | 2 | 0.5 | 3.5

Core • 13-22 • 2.65 | 9.93 | 4.75 | 6.32 | 3.55 | 2.55 | 9.91 | 4.65 | 3.55 | 7.55

Challenge • 23-26 • 4.544 | 1.65 | 5.554 | 2.35

1.3 • Multiplying Decimals

OUT OF 22

Warm-Up • 1-12 • 2 | 1.2 | 3 | 1 | 0.3 | 5 | 3 | 1.6 | 6 | 2 | 5 | 0.9

Core • 13-22 • 0.08 | 0.6 | 3.5 | 2.4 | 8 | 0.12 | 0.9 | 4.2 | 4 | 8.4

Challenge • 23-26 • 1 | 1 | 3 | 3

1.4 • Dividing Decimals

OUT OF 22

Warm-Up • 1-12 • 2.4 | 0.3 | 1.6 | 0.5 | 0.06 | 0.9 | 3.2 | 0.2 | 0.9 | 16 | 0.4 | 24

Core • 13-22 • 12 | 16 | 3 | 8 | 2.5 | 1.2 | 0.32 | 30 | 8 | 0.8

Challenge • 23-26 • 1.2 | 6 | 3.5 | 125

Fri • Unit Price Investigation

OUT OF 8

Warm-Up • 1-4 • 3 | 2 | 3 | 2

Core • 5-8 • 1.5 | 9 | 1.35 | 1.45

Challenge • 9-14 • 1.125 | 1.25 | B | 1.2 | 1.1 | 1kg

Ignore the Point

Multiply the digits first, then count the places.

Warm-Up 12 ITEMS • RECALL

1 5×4

2 0.5×4

3 12×5

4 1.2×5

5 25×4

6 0.25×4

7 6×3

8 0.6×3

9 14×5

10 1.4×5

11 35×4

12 0.35×4

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 0.4×0.2

14. 1.2×0.5

15. 2.5×1.4

16. 0.06×40

17. 3.2×2.5

18. 0.6×0.3

19. 1.4×0.5

20. 3.5×1.6

21. 0.07×30

22. 4.2×2.5

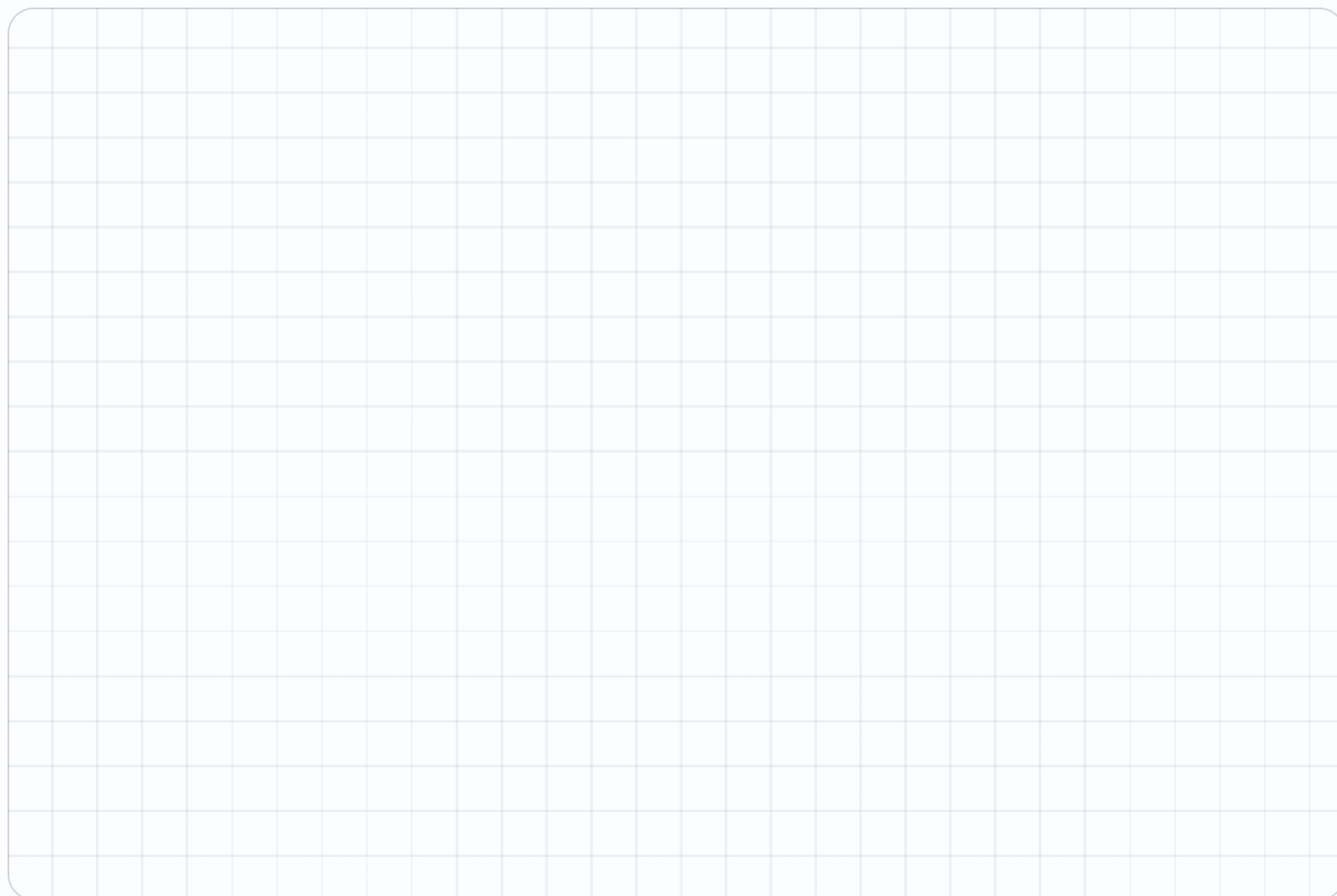
★ Challenge NOT GRADED • REASON IT OUT

23. 0.125×8
.....

24. 1.25×0.8
.....

25. 0.375×16
.....

26. 2.25×0.8
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Count the Places

Two decimal digits in, two decimal digits out.

Warm-Up 12 ITEMS • RECALL

1 Decimal digits in 0.4

2 In 0.25

3 In 1.234

4 0.2×0.3

5 0.5×0.2

6 0.1×0.1

7 Decimal digits in 0.06

8 In 3.5

9 In 0.4567

10 0.3×0.3

11 0.4×0.5

12 0.2×0.2

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 0.25×0.4

14. 1.5×0.02

15. 0.12×0.5

16. 2.4×0.25

17. 0.75×0.8

18. 0.35×0.2

19. 2.5×0.04

20. 0.14×0.5

21. 3.6×0.25

22. 0.85×0.4

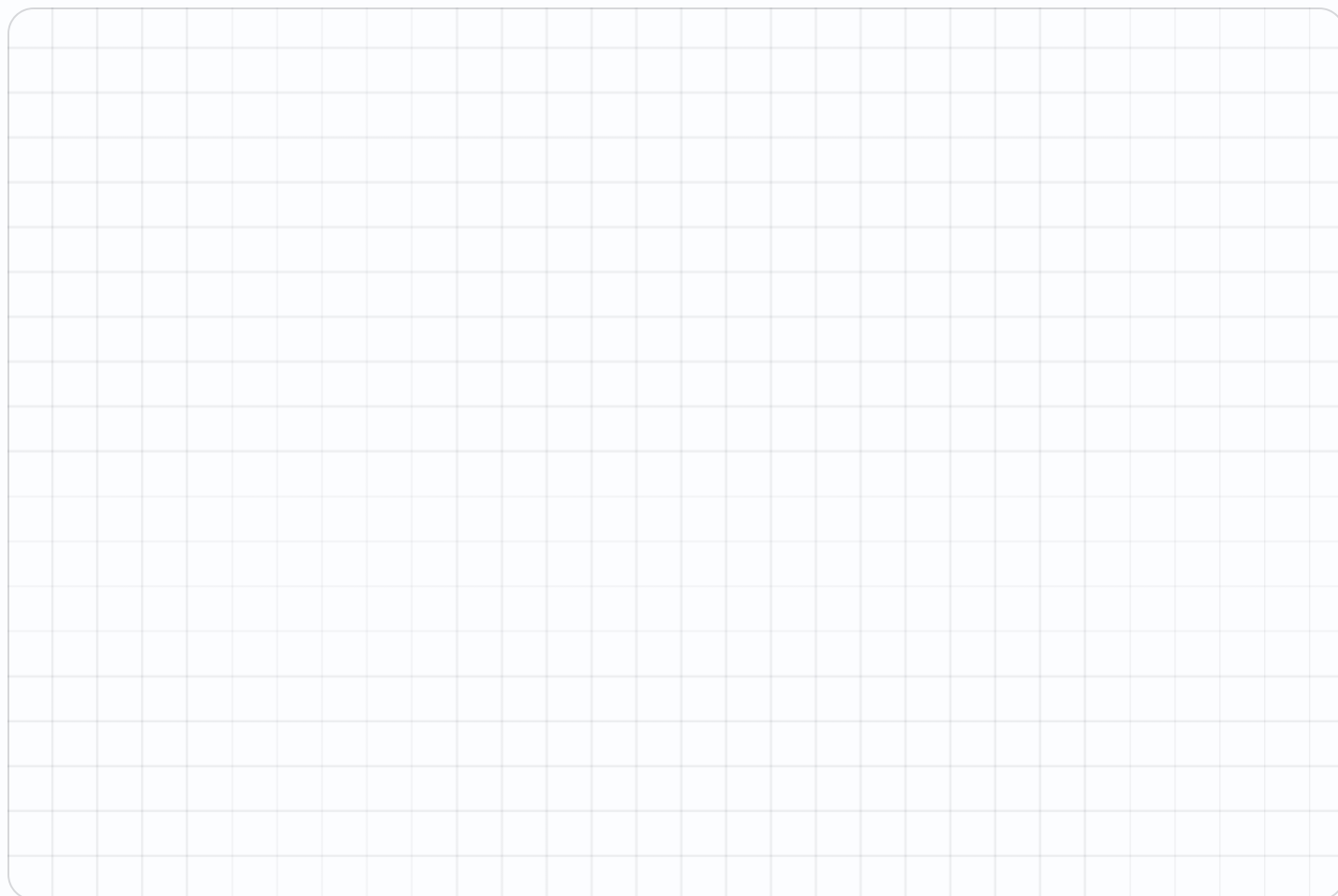
★ Challenge NOT GRADED • REASON IT OUT

23. 0.125×0.4
.....

24. 1.25×1.6
.....

25. 0.375×0.8
.....

26. 1.75×1.6
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Place It by Estimating

0.4×60 is about 24, so the point can only go one place.

Warm-Up 12 ITEMS • RECALL

1 Estimate 0.5×40

2 0.5×40

3 Estimate 2×30

4 1.9×30

5 Estimate 0.25×80

6 0.25×80

7 Estimate 0.5×60

8 0.5×60

9 Estimate 3×20

10 2.9×20

11 Estimate 0.25×40

12 0.25×40

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Estimate 0.4×60

.....

14. 0.4×62

.....

15. Estimate 4.9×21

.....

16. 4.9×20

.....

17. Estimate 0.9×150

.....

18. Estimate 0.6×80

.....

19. 0.6×82

.....

20. Estimate 3.9×31

.....

21. 3.9×30

.....

22. Estimate 0.8×250

.....

Smaller Than You Started

Why $\times 0.5$ halves instead of doubling.

Warm-Up 12 ITEMS • RECALL

1 10×0.5

2 10×2

3 10×1

4 8×0.5

5 20×0.25

6 6×0.5

7 20×0.5

8 20×2

9 12×0.5

10 40×0.25

11 20×1

12 16×0.5

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 40×0.75 — bigger or smaller than 40

14. 40×0.75

15. 40×1.25 — bigger or smaller

16. 40×1.25

17. 100×0.1

18. 60×0.75 — bigger or smaller than 60

19. 60×0.75

20. 60×1.5 — bigger or smaller

21. 60×1.5

22. 200×0.1



Point Placement

Given the digits, race to place the decimal point.

✦ Warm-Up 4 ITEMS • RECALL

1 Digits 24, one decimal place — the number

2 Digits 24, two places

3 Digits 36, one decimal place — the number

4 Digits 36, two places

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. 0.6×0.4 — digits 24, so the answer

.....

6. 6×0.4 — the answer

.....

7. 0.9×0.4 — digits 36, so the answer

.....

8. 9×0.4 — the answer

.....

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 • Ignore the Point

OUT OF 22

Warm-Up • 1-12 • 20 | 2 | 60 | 6 | 100 | 1 | 18 | 1.8 | 70 | 7 | 140 | 1.4

Core • 13-22 • 0.08 | 0.6 | 3.5 | 2.4 | 8 | 0.18 | 0.7 | 5.6 | 2.1 | 10.5

Challenge • 23-26 • 1 | 1 | 6 | 1.8

2.2 • Count the Places

OUT OF 22

Warm-Up • 1-12 • 1 | 2 | 3 | 0.06 | 0.1 | 0.01 | 2 | 1 | 4 | 0.09 | 0.2 | 0.04

Core • 13-22 • 0.1 | 0.03 | 0.06 | 0.6 | 0.6 | 0.07 | 0.1 | 0.07 | 0.9 | 0.34

Challenge • 23-26 • 0.05 | 2 | 0.3 | 2.8

2.3 • Place It by Estimating

OUT OF 22

Warm-Up • 1-12 • 20 | 20 | 60 | 57 | 20 | 20 | 30 | 30 | 60 | 58 | 10 | 10

Core • 13-22 • 24 | 24.8 | 100 | 98 | 135 | 48 | 49.2 | 120 | 117 | 200

Challenge • 23-26 • 180 | 500 | 240 | 900

2.4 • Smaller Than You Started

OUT OF 22

Warm-Up • 1-12 • 5 | 20 | 10 | 4 | 5 | 3 | 10 | 40 | 6 | 10 | 20 | 8

Core • 13-22 • smaller | 30 | bigger | 50 | 10 | smaller | 45 | bigger | 90 | 20

Challenge • 23-26 • smaller | 3 | same | 4

Fri • Point Placement

OUT OF 8

Warm-Up • 1-4 • 2.4 | 0.24 | 3.6 | 0.36

Core • 5-8 • 0.24 | 2.4 | 0.36 | 3.6

Challenge • 9-14 • 0.24 | 0.24 | 0.24 | 0.36 | 0.36 | 0.36

Decimal ÷ Whole

The point comes straight up. Nothing else moves.

Warm-Up 12 ITEMS • RECALL

1 $4.8 \div 2$

2 $0.9 \div 3$

3 $6.4 \div 4$

4 $2.5 \div 5$

5 $0.36 \div 6$

6 $7.2 \div 8$

7 $6.9 \div 3$

8 $0.8 \div 2$

9 $8.4 \div 4$

10 $3.5 \div 7$

11 $0.48 \div 8$

12 $5.4 \div 9$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $9.6 \div 4$

14. $1.44 \div 12$

15. $5.25 \div 5$

16. $8.4 \div 7$

17. $0.72 \div 9$

18. $1.32 \div 11$

19. $6.35 \div 5$

20. $9.8 \div 7$

21. $0.63 \div 9$



Whole ÷ Decimal

Shift both the same number of places, then divide as usual.

Warm-Up 11 ITEMS • RECALL

1 $6 \div 0.5$

2 $4 \div 0.5$

3 $10 \div 0.5$

4 $4 \div 0.25$

5 $3 \div 0.5$

6 $2 \div 0.25$

7 $8 \div 0.5$

8 $20 \div 0.5$

9 $8 \div 0.25$

10 $5 \div 0.5$

11 $3 \div 0.25$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. $9 \div 0.3$

13. $12 \div 0.4$

14. $5 \div 0.125$

15. $7 \div 0.7$

16. $20 \div 0.8$

17. $12 \div 0.3$

18. $18 \div 0.6$

19. $6 \div 0.125$

20. $9 \div 0.9$

21. $30 \div 0.8$

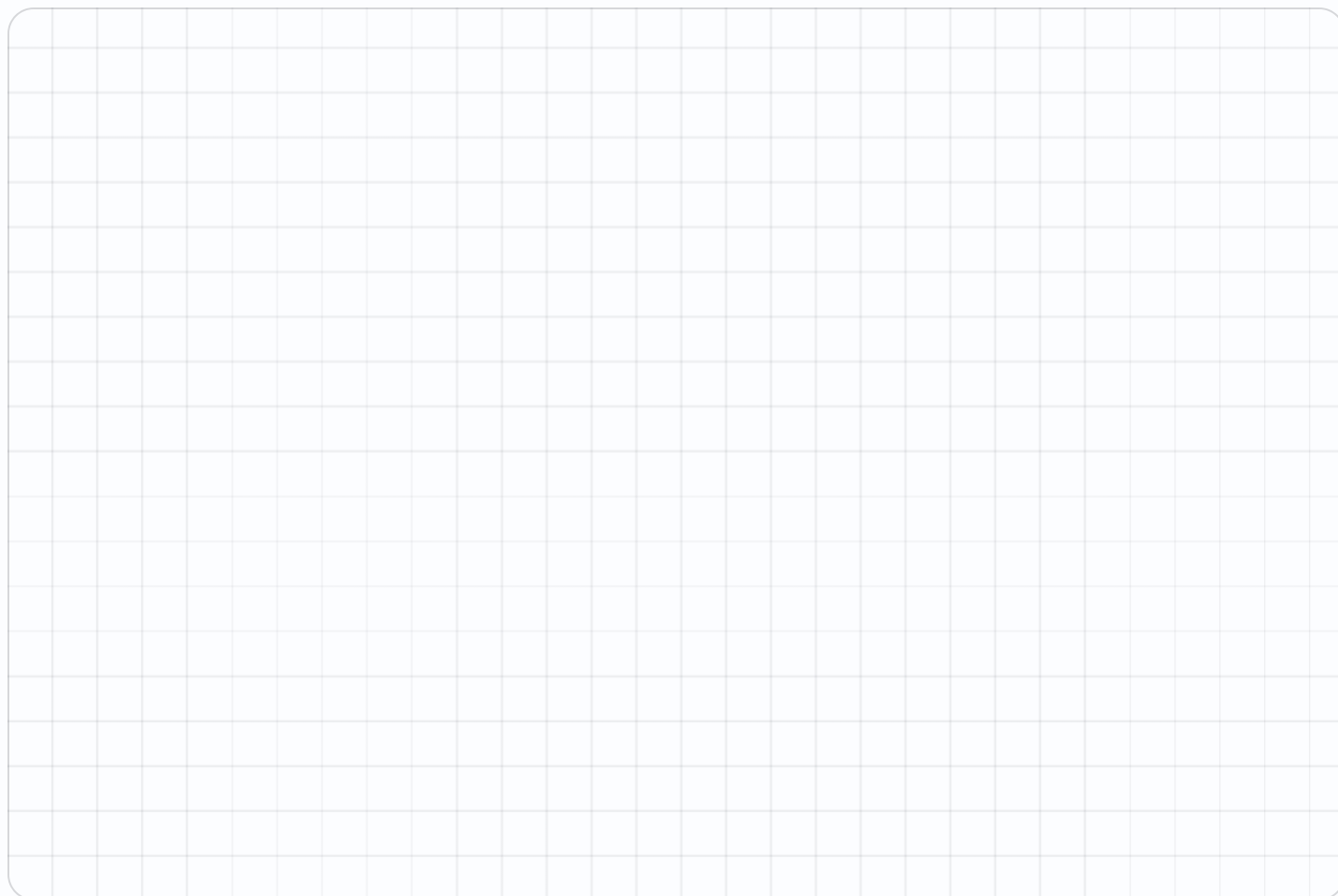
★ Challenge NOT GRADED • REASON IT OUT

22. $15 \div 0.06$
.....

23. $1 \div 0.004$
.....

24. $21 \div 0.07$
.....

25. $2 \div 0.004$
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Decimal ÷ Decimal

Same shift, applied to both numbers.

✦ Warm-Up 11 ITEMS • RECALL

1 $0.6 \div 0.2$

2 $0.8 \div 0.4$

3 $1.5 \div 0.5$

4 $0.9 \div 0.3$

5 $2.4 \div 0.6$

6 $1.2 \div 0.4$

7 $1.2 \div 0.6$

8 $2.5 \div 0.5$

9 $1.8 \div 0.3$

10 $3.6 \div 0.9$

11 $1.6 \div 0.4$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. $9.6 \div 1.2$

13. $0.144 \div 0.12$

14. $4.5 \div 1.5$

15. $6.25 \div 2.5$

16. $0.81 \div 0.09$

17. $8.4 \div 1.2$

18. $0.288 \div 0.12$

19. $6.3 \div 1.5$

20. $7.5 \div 2.5$

21. $0.64 \div 0.08$



Challenge

22. $7.5 \div 0.25$

23. $0.0144 \div 0.012$

24. $9.6 \div 0.32$

25. $0.0288 \div 0.012$



Working space



Error journal

Money Problems

Unit prices and change, to the cent.

Warm-Up 12 ITEMS • RECALL

1 \$6 for 2 — each

2 \$10 for 4 — each

3 \$9 for 3 — each

4 \$20 — \$12.50

5 \$5 ÷ 4

6 3 at \$2.50

7 \$8 for 2 — each

8 \$12 for 4 — each

9 \$15 for 3 — each

10 \$20 — \$14.50

11 \$6 ÷ 4

12 4 at \$2.25

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. \$7.50 shared by 3

.....

14. \$4.50 for 3 kg — per kg

.....

15. \$7.20 for 0.8 kg — per kg

.....

16. \$20 — 3 at \$4.99

.....

17. \$15 ÷ 12

.....

18. \$9.60 shared by 4

.....

19. \$6.75 for 3 kg — per kg

.....

20. \$8.40 for 0.6 kg — per kg

.....

21. \$50 — 4 at \$6.99

.....

22. \$18 ÷ 12

.....



Working space



Error journal

Unit Price Investigation Begins

Collect real prices and sizes. No conclusions yet.

✦ Warm-Up 4 ITEMS • RECALL

1 \$8 for 4 — each

2 \$12 for 6 — each

3 \$10 for 5 — each

4 \$18 for 6 — each

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. \$3.60 for 4 — each

.....

6. \$11.25 for 5 — each

.....

7. \$4.80 for 4 — each

.....

8. \$13.75 for 5 — each

.....

Challenge NOT GRADED • REASON IT OUT

9. 1 kg at \$8, or 750 g at \$5.40 — per kg for the small one

10. Which is better value — type 1kg or 750g

11. \$2.40 for 6 — each

12. 2 kg at \$15, or 1.5 kg at \$12 — per kg for the small one

13. Which is better value — type 2kg or 1.5kg

14. \$3.60 for 8 — each

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 3 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

3.1 • Decimal ÷ Whole

OUT OF 21

Warm-Up • 1-12 • 2.4 | 0.3 | 1.6 | 0.5 | 0.06 | 0.9 | 2.3 | 0.4 | 2.1 | 0.5 | 0.06
| 0.6

Core • 13-21 • 2.4 | 0.12 | 1.05 | 1.2 | 0.08 | 0.12 | 1.27 | 1.4 | 0.07

Challenge • 22-25 • 3.125 | 0.49 | 3.625 | 0.63

3.2 • Whole ÷ Decimal

OUT OF 21

Warm-Up • 1-11 • 12 | 8 | 20 | 16 | 6 | 8 | 16 | 40 | 32 | 10 | 12

Core • 12-21 • 30 | 30 | 40 | 10 | 25 | 40 | 30 | 48 | 10 | 37.5

Challenge • 22-25 • 250 | 250 | 300 | 500

3.3 • Decimal ÷ Decimal

OUT OF 21

Warm-Up • 1-11 • 3 | 2 | 3 | 3 | 4 | 3 | 2 | 5 | 6 | 4 | 4

Core • 12-21 • 8 | 1.2 | 3 | 2.5 | 9 | 7 | 2.4 | 4.2 | 3 | 8

Challenge • 22-25 • 30 | 1.2 | 30 | 2.4

3.4 • Money Problems

OUT OF 22

Warm-Up • 1-12 • 3 | 2.5 | 3 | 7.5 | 1.25 | 7.5 | 4 | 3 | 5 | 5.5 | 1.5 | 9

Core • 13-22 • 2.5 | 1.5 | 9 | 5.03 | 1.25 | 2.4 | 2.25 | 14 | 22.04 | 1.5

Challenge • 23-26 • 1.25 | 1.125 | 1.2 | 1.1

Fri • Unit Price Investigation Begins

OUT OF 8

Warm-Up • 1-4 • 2 | 2 | 2 | 3

Core • 5-8 • 0.9 | 2.25 | 1.2 | 2.75

Challenge • 9-14 • 7.2 | 750g | 0.4 | 8 | 2kg | 0.45

Choose the Operation

The words tell you. Read them twice before reaching for a method.

Warm-Up 12 ITEMS • RECALL

1 3 items at \$2.50 — total

2 \$10 shared by 4

3 \$5.50 + \$2.50

4 \$10 – \$3.25

5 4 at \$1.25

6 \$9 ÷ 3

7 4 items at \$3.50 — total

8 \$12 shared by 3

9 \$6.50 + \$3.50

10 \$20 – \$4.75

11 5 at \$1.40

12 \$18 ÷ 6

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 2.5 kg at \$4 per kg

.....

14. \$12 buys how many at \$2.40

.....

15. 1.5 m plus 0.85 m — metres

.....

16. 6 at \$3.99 — total

.....

17. \$50 – \$27.45

.....

18. 3.5 kg at \$6 per kg

.....

19. \$15 buys how many at \$2.50

.....

20. 2.4 m plus 1.35 m — metres

.....

21. 7 at \$2.99 — total

.....

22. \$80 – \$34.65

.....

Two-Step Problems

Multiply then subtract, in that order.

Warm-Up 10 ITEMS • RECALL

1 3 at \$2, change from \$10

2 4 at \$1.50, change from \$10

3 2 at \$3.25 — total

4 Change from \$10

5 5 at \$2 — total

6 Change from \$20

7 4 at \$2, change from \$10

8 3 at \$1.50, change from \$10

9 3 at \$2.25 — total

10 6 at \$2 — total

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. 3 at \$4.99, change from \$20

.....

12. 6 at \$1.75 — total

.....

13. 2.5 kg at \$3.20, change from \$10

.....

14. 4 at \$6.25 — total

.....

15. 4 at \$3.99, change from \$20

.....

16. 5 at \$1.85 — total

.....

17. 3.5 kg at \$2.40, change from \$10

.....

18. 3 at \$8.25 — total

.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Measurement Contexts

Metres, litres and kilograms, all carrying decimals.

✦ Warm-Up 12 ITEMS • RECALL

1 $1.5 \text{ m} + 0.5 \text{ m}$

2 $2.5 \text{ L} - 1 \text{ L}$

3 $0.5 \text{ kg} \times 4$

4 $3 \text{ m} \div 2$

5 $1.2 \text{ m} + 0.8 \text{ m}$

6 $4.5 \text{ kg} - 1.5 \text{ kg}$

7 $2.5 \text{ m} + 1.5 \text{ m}$

8 $3.5 \text{ L} - 1.5 \text{ L}$

9 $0.25 \text{ kg} \times 8$

10 $5 \text{ m} \div 2$

11 $1.6 \text{ m} + 0.4 \text{ m}$

12 $6.5 \text{ kg} - 2.5 \text{ kg}$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $1.25 \text{ m} \times 4$

14. 7.5 L shared into 0.5 L bottles — how many

15. $2.4 \text{ kg} \div 3$

16. $0.75 \text{ m} \times 6$

17. $12.5 \text{ L} - 4.75 \text{ L}$

18. $1.75 \text{ m} \times 4$

19. 9 L shared into 0.75 L bottles — how many

20. $3.6 \text{ kg} \div 4$

21. $0.45 \text{ m} \times 8$

22. $20.5 \text{ L} - 8.75 \text{ L}$



Working space



Error journal

Estimate Everything First

Write the estimate before the working, every time.

Warm-Up 12 ITEMS • RECALL

1 Estimate $4.9 + 3.1$

2 Estimate $9.8 - 4.9$

3 Estimate 2.1×4

4 Estimate $11.9 \div 4$

5 Estimate $0.9 + 1.1$

6 Estimate 5.2×2

7 Estimate $5.9 + 4.1$

8 Estimate $10.2 - 5.1$

9 Estimate 3.1×3

10 Estimate $15.9 \div 4$

11 Estimate $1.9 + 2.1$

12 Estimate 6.1×2

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Estimate $12.4 - 7.9$

.....

14. True value of $12.4 - 7.856$

.....

15. Estimate 0.49×100

.....

16. True value of 0.49×100

.....

17. Estimate $19.8 \div 5$

.....

18. Estimate $14.6 - 8.9$

.....

19. True value of $14.6 - 8.746$

.....

20. Estimate 0.51×200

.....

21. True value of 0.51×200

.....

22. Estimate $24.6 \div 5$

.....



Mid-Unit Quiz

Eight items across Weeks 1–4. 85% to keep flying.

✦ Warm-Up 4 ITEMS • RECALL

1 $0.3 + 0.4$

2 0.5×4

3 $0.5 + 0.2$

4 0.6×3

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. $3.5 + 0.47$

6. $8.2 - 3.45$

7. 0.4×0.2

8. $6 \div 0.5$

9. \$7.20 for 0.8 kg — per kg

10. $4.6 + 0.38$

11. $9.2 - 4.55$

12. 0.6×0.3

13. $8 \div 0.5$

14. \$8.40 for 0.6 kg — per kg

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Choose the Operation

OUT OF 22

Warm-Up • 1-12 • 7.5 | 2.5 | 8 | 6.75 | 5 | 3 | 14 | 4 | 10 | 15.25 | 7 | 3

Core • 13-22 • 10 | 5 | 2.35 | 23.94 | 22.55 | 21 | 6 | 3.75 | 20.93 | 45.35

Challenge • 23-26 • 4.8 | 24 | 9 | 60

4.2 • Two-Step Problems

OUT OF 18

Warm-Up • 1-10 • 4 | 4 | 6.5 | 3.5 | 10 | 10 | 2 | 5.5 | 6.75 | 12

Core • 11-18 • 5.03 | 10.5 | 2 | 25 | 4.04 | 9.25 | 1.6 | 24.75

Challenge • 19-22 • 6.25 | 9.7 | 9 | 17.5

4.3 • Measurement Contexts

OUT OF 22

Warm-Up • 1-12 • 2 | 1.5 | 2 | 1.5 | 2 | 3 | 4 | 2 | 2 | 2.5 | 2 | 4

Core • 13-22 • 5 | 15 | 0.8 | 4.5 | 7.75 | 7 | 12 | 0.9 | 3.6 | 11.75

Challenge • 23-26 • 6 | 0.45 | 7 | 0.45

4.4 • Estimate Everything First

OUT OF 22

Warm-Up • 1-12 • 8 | 5 | 8 | 3 | 2 | 10 | 10 | 5 | 9 | 4 | 4 | 12

Core • 13-22 • 4 | 4.544 | 50 | 49 | 4 | 6 | 5.854 | 100 | 102 | 5

Challenge • 23-26 • 30 | 29.7 | 32 | 31.6

Fri • Mid-Unit Quiz

OUT OF 14

Warm-Up • 1-4 • 0.7 | 2 | 0.7 | 1.8

Core • 5-14 • 3.97 | 4.75 | 0.08 | 12 | 9 | 4.98 | 4.65 | 0.18 | 16 | 14

Challenge • 15-16 • 4.544 | 5.554

Compute Unit Prices

Divide price by size for every item on the list.

Warm-Up 12 ITEMS • RECALL

1 \$6 for 2 kg — per kg

2 \$10 for 5 L — per L

3 \$8 for 4 — each

4 \$12 for 3 — each

5 \$15 for 5 — each

6 \$9 for 3 — each

7 \$8 for 2 kg — per kg

8 \$12 for 6 L — per L

9 \$10 for 5 — each

10 \$16 for 4 — each

11 \$18 for 6 — each

12 \$21 for 7 — each

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. \$4.50 for 3 kg — per kg

.....

14. \$7.20 for 0.8 kg — per kg

.....

15. \$11.25 for 5 — each

.....

16. \$2.40 for 6 — each

.....

17. \$13.50 for 1.5 kg — per kg

.....

18. \$6.75 for 3 kg — per kg

.....

19. \$8.40 for 0.6 kg — per kg

.....

20. \$13.75 for 5 — each

.....

21. \$3.60 for 8 — each

.....

22. \$16.20 for 1.8 kg — per kg

.....



Working space



Error journal

Find the Trap

Name one product where the bigger box costs more per unit.

Warm-Up 10 ITEMS • RECALL

1 \$4 for 2 — each

2 \$9 for 5 — each

3 Cheaper each — type 1.8 or 2

4 \$6 for 3 — each

5 \$10 for 4 — each

6 Cheaper each — type 2 or 2.5

7 \$10 for 6 — each, to the cent

8 Cheaper each — type 1.67 or 2

9 \$8 for 4 — each

10 \$15 for 6 — each

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. 400 g at \$5 — per 100 g

.....

12. 800 g at \$9 — per 100 g

.....

13. Better value — type 400 or 800

.....

14. 500 g at \$4 — per 100 g

.....

15. 1 kg at \$9 — per 100 g

.....

16. 500 g at \$6 — per 100 g

.....

17. 1 kg at \$11 — per 100 g

.....

18. Better value — type 500 or 1000

.....

19. 250 g at \$3 — per 100 g

.....

20. 750 g at \$8.25 — per 100 g

.....

Write the Recommendation

One paragraph a shopper could act on.

Warm-Up 12 ITEMS • RECALL

1 \$1.25 per 100 g — the cost of 400 g

2 \$0.90 per 100 g — the cost of 1 kg

3 \$2 each — the cost of 5

4 \$1.80 each — the cost of 5

5 The saving

6 \$3 per kg — the cost of 2 kg

7 \$1.20 per 100 g — the cost of 500 g

8 \$1.10 per 100 g — the cost of 1 kg

9 \$2 each — the cost of 6

10 \$1.67 each — the cost of 6, to the cent

11 \$3 per kg — the cost of 3 kg

12 \$4 each — the cost of 4

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Saving \$0.125 per 100 g on 800 g

.....

14. Buying 800 g weekly — the yearly saving over 52 weeks

.....

15. \$1.25 vs \$1.125 per 100 g — the percent saved, to the nearest whole

.....

16. \$9 vs \$10 — the percent saved

.....

17. Saving \$1 a week for a year

.....

18. Saving \$0.10 per 100 g on 1 kg

.....

19. Buying 1 kg weekly — the yearly saving over 52 weeks

.....

20. \$1.20 vs \$1.10 per 100 g — the percent saved, to the nearest whole

.....

21. \$11 vs \$12 — the percent saved, to the nearest whole

.....

22. Saving \$2 a week for a year

.....

 Challenge NOT GRADED • REASON IT OUT

23. A \$0.10 per 100 g saving on 2 kg weekly, over 52 weeks

24. Which matters more for a weekly buy — type per-unit or total

25. A \$0.15 per 100 g saving on 3 kg weekly, over 52 weeks

26. Which number belongs in the headline — type per-unit or total

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

Warm-Up 12 ITEMS • RECALL

1 $0.25 + 0.25$

2 $1 - 0.5$

3 0.5×4

4 $4.8 \div 2$

5 $0.6 + 0.9$

6 \$10 for 5 — each

7 $0.35 + 0.15$

8 $1 - 0.4$

9 0.2×8

10 $6.9 \div 3$

11 $0.8 + 0.7$

12 \$12 for 4 — each

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $12.6 + 4.85$

14. $10 - 0.07$

15. 0.4×0.2

16. $9.6 \div 1.2$

17. 2.5×1.4

18. $15.4 + 3.75$

19. $10 - 0.09$

20. 0.6×0.3

21. $8.4 \div 1.2$

22. 3.5×1.6

Mission 04 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 4 ITEMS • RECALL

1 $1.2 + 2.5$

2 $0.9 - 0.4$

3 $1.4 + 2.3$

4 $0.8 - 0.5$

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. $3.5 + 0.47$

6. $4 - 1.35$

7. 1.2×0.5

8. 3.2×2.5

9. $6 \div 0.5$

10. $9.6 \div 1.2$

11. \$4.50 for 3 kg — per kg

12. 3 at \$4.99 — change from \$20

13. $4.6 + 0.38$

14. $5 - 2.45$

15. 1.4×0.5

16. 4.2×2.5

17. $8 \div 0.5$

18. $8.4 \div 1.2$

19. \$6.75 for 3 kg — per kg

20. 4 at \$3.99 — change from \$20



Week 5 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

5.1 • Compute Unit Prices

OUT OF 22

Warm-Up • 1-12 • 3 | 2 | 2 | 4 | 3 | 3 | 4 | 2 | 2 | 4 | 3 | 3

Core • 13-22 • 1.5 | 9 | 2.25 | 0.4 | 9 | 2.25 | 14 | 2.75 | 0.45 | 9

Challenge • 23-26 • 1.25 | 1.125 | 1.2 | 1.1

5.2 • Find the Trap

OUT OF 20

Warm-Up • 1-10 • 2 | 1.8 | 1.8 | 2 | 2.5 | 2 | 1.67 | 1.67 | 2 | 2.5

Core • 11-20 • 1.25 | 1.125 | 800 | 0.8 | 0.9 | 1.2 | 1.1 | 1000 | 1.2 | 1.1

Challenge • 21-23 • 500g | 0.1 | 750g

5.3 • Write the Recommendation

OUT OF 22

Warm-Up • 1-12 • 5 | 9 | 10 | 9 | 1 | 6 | 6 | 11 | 12 | 10.02 | 9 | 16

Core • 13-22 • 1 | 52 | 10 | 10 | 52 | 1 | 52 | 8 | 8 | 104

Challenge • 23-26 • 104 | per-unit | 234 | per-unit

Thu • Error Journal Sweep

OUT OF 22

Warm-Up • 1-12 • 0.5 | 0.5 | 2 | 2.4 | 1.5 | 2 | 0.5 | 0.6 | 1.6 | 2.3 | 1.5 | 3

Core • 13-22 • 17.45 | 9.93 | 0.08 | 8 | 3.5 | 19.15 | 9.91 | 0.18 | 7 | 5.6

Challenge • 23-26 • 4.544 | 1 | 5.554 | 1.8

Fri • Mission 04 Test

OUT OF 20

Warm-Up • 1-4 • 3.7 | 0.5 | 3.7 | 0.3

Core • 5-20 • 3.97 | 2.65 | 0.6 | 8 | 12 | 8 | 1.5 | 5.03 | 4.98 | 2.55 | 0.7 |
10.5 | 16 | 7 | 2.25 | 4.04

Challenge • 21-24 • 4.544 | 6 | 5.554 | 7